

Analysis group meeting

Mathieu Ouillon

5/6/25

- Pass0v11 (40 files per run):
 - Cooking is done.
 - Detector timelines are done.
 - High-level physics QA timelines are also done.
- Done with COATJAVA 12.0.5t, new network for the forward tracking and denoising.
- I started to skim the data yesterday, but got an issue (core dump).
- I have restarted.

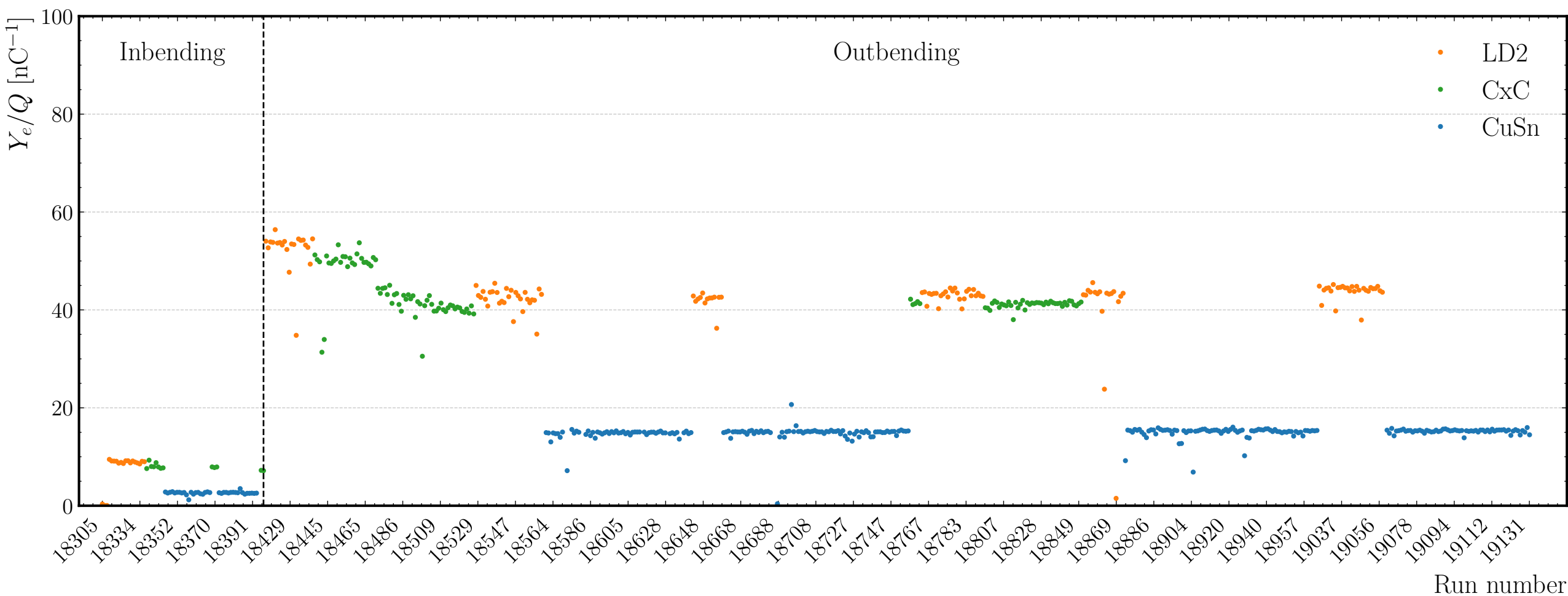
- Total events: 40,164,051,428
 - Total number of EVIO files: 573,493
 - Number of runs: 619
 - Number of events per EVIO file: approx. 74,000
-
- Size of raw file (2GB) 1146 TB on mss ($2\text{GB} \times 573,493$)
 - Size of decoded files: 442TB decoded pass0v11: 19.1TB for 619*40 files, for 573,493 →
573,493*19.1/(619*40)
 - Size of DST files: 172TB based on run 18542 → 12GB → $573,493 \times 12 / 40 = 172000\text{GB}$

- Size skim forward: 11:X+:X-:Xn
 - For 40 files: 10GB
 - Total: 143TB
- Size skim forward: 11:X+:X-:Xn && $Q^2 > 0.7 \text{ GeV}^2$ && $W > 1.8 \text{ GeV}$
 - For 40 files: 6.9GB
 - Total: 98.9TB
- Size JPSITCS:
 - For 40 files: 190MB
 - Total: 2.7TB

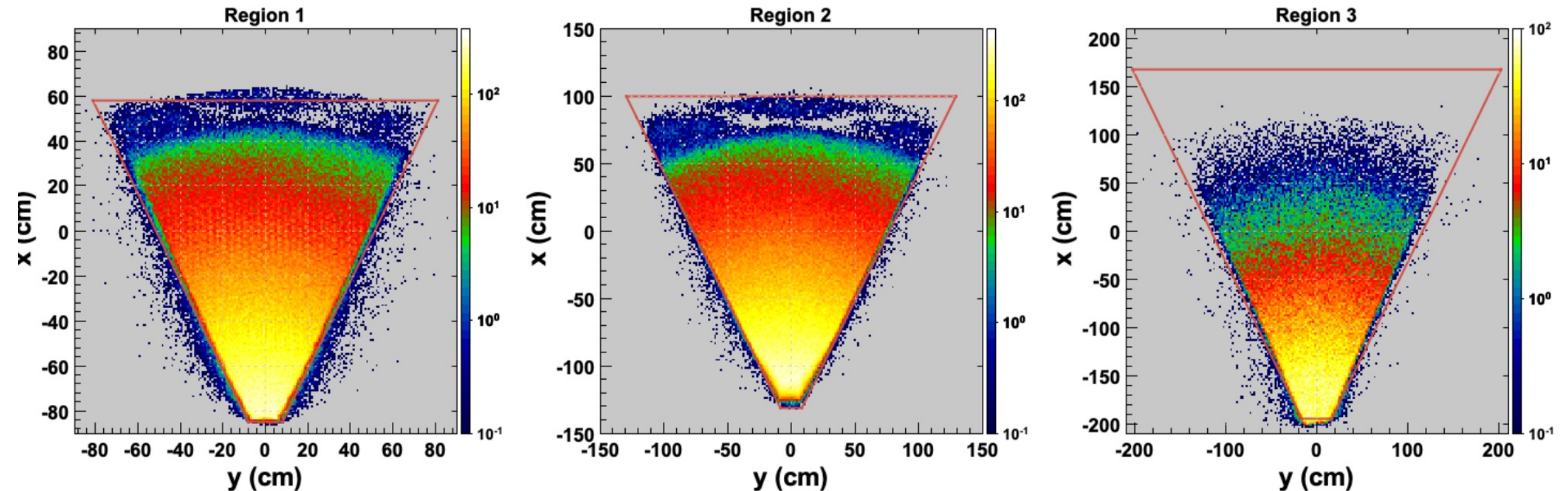
- Try to fix the electron FD normalized yield timeline:
- Instead of dividing by the total charged for the run, divide by the charge normalized by the number of files cooked:

$$Y_e = \frac{Y_e}{Q \frac{\text{number of files cooked}}{\text{total number of files per run}}}$$

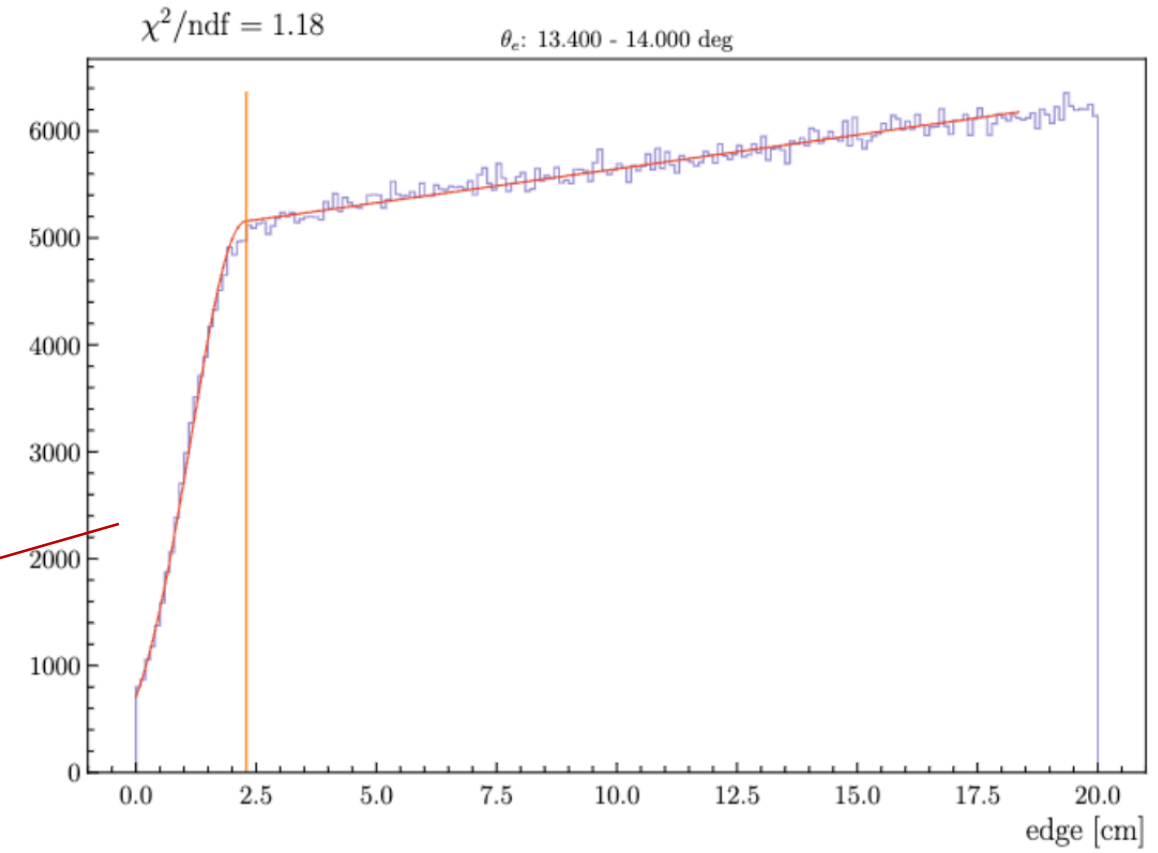
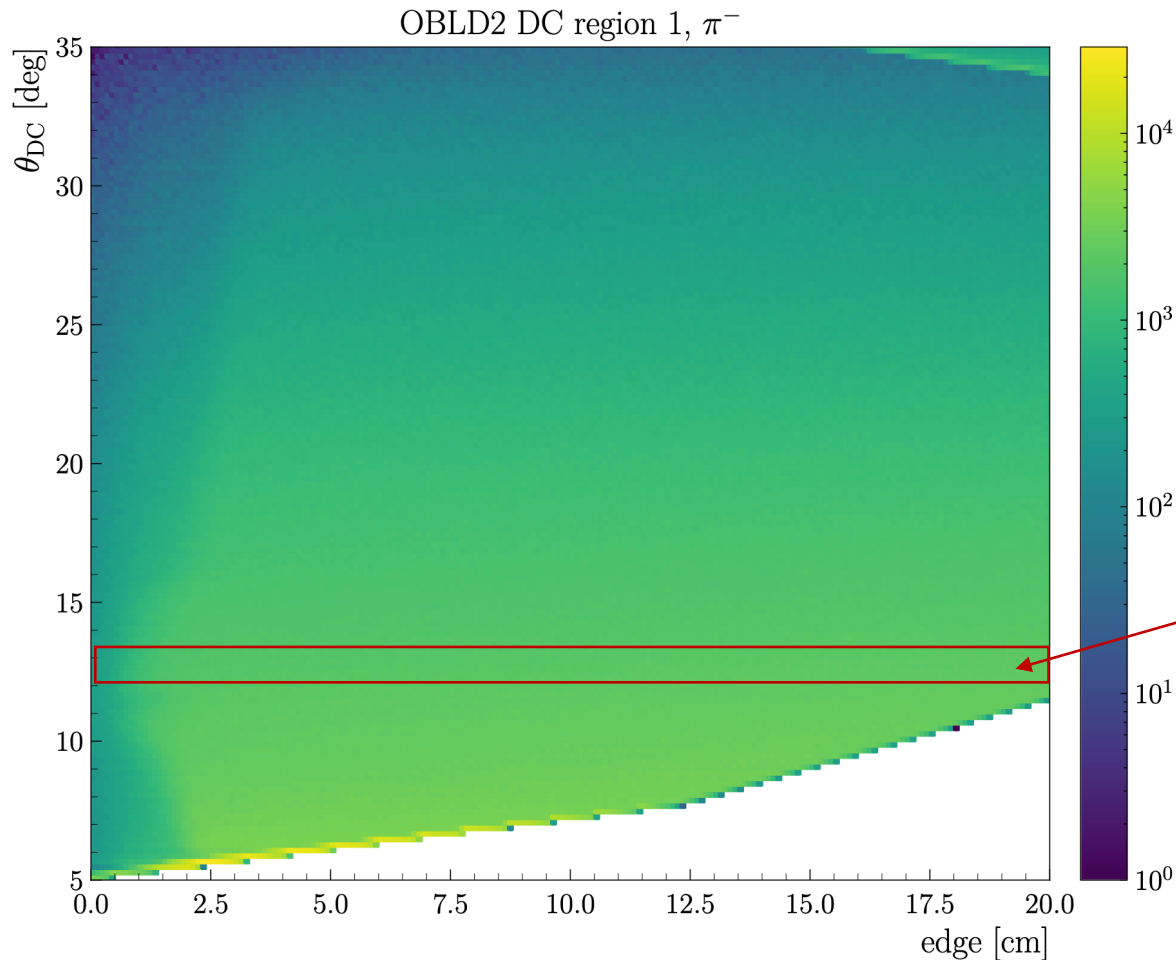
- Try to fix the electron FD normalized yield timeline:
- Instead of dividing by the total charged for the run, divide by the charge normalized by the number of files cooked:



- Switch from using χ^2/ndf and edge to theta and edge based on the [study made by Raffaella](#).
- REC::Traj.edge, distance between the trajectory point and the detector edge
- Edge is based on the geometry package (it is the edge of the DC).

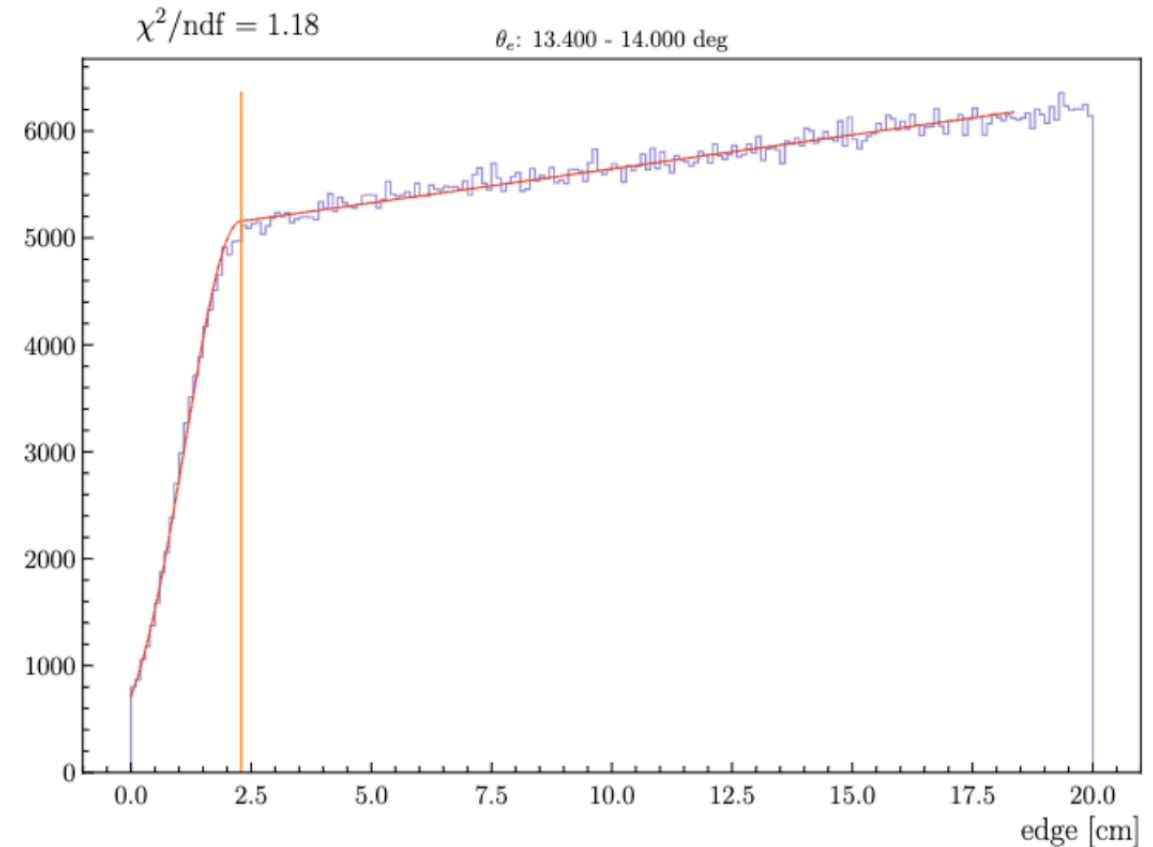


- Defining the “edge” limit
- Theta coordinates in the tilted-sector frame (all the sectors are on top of each other)

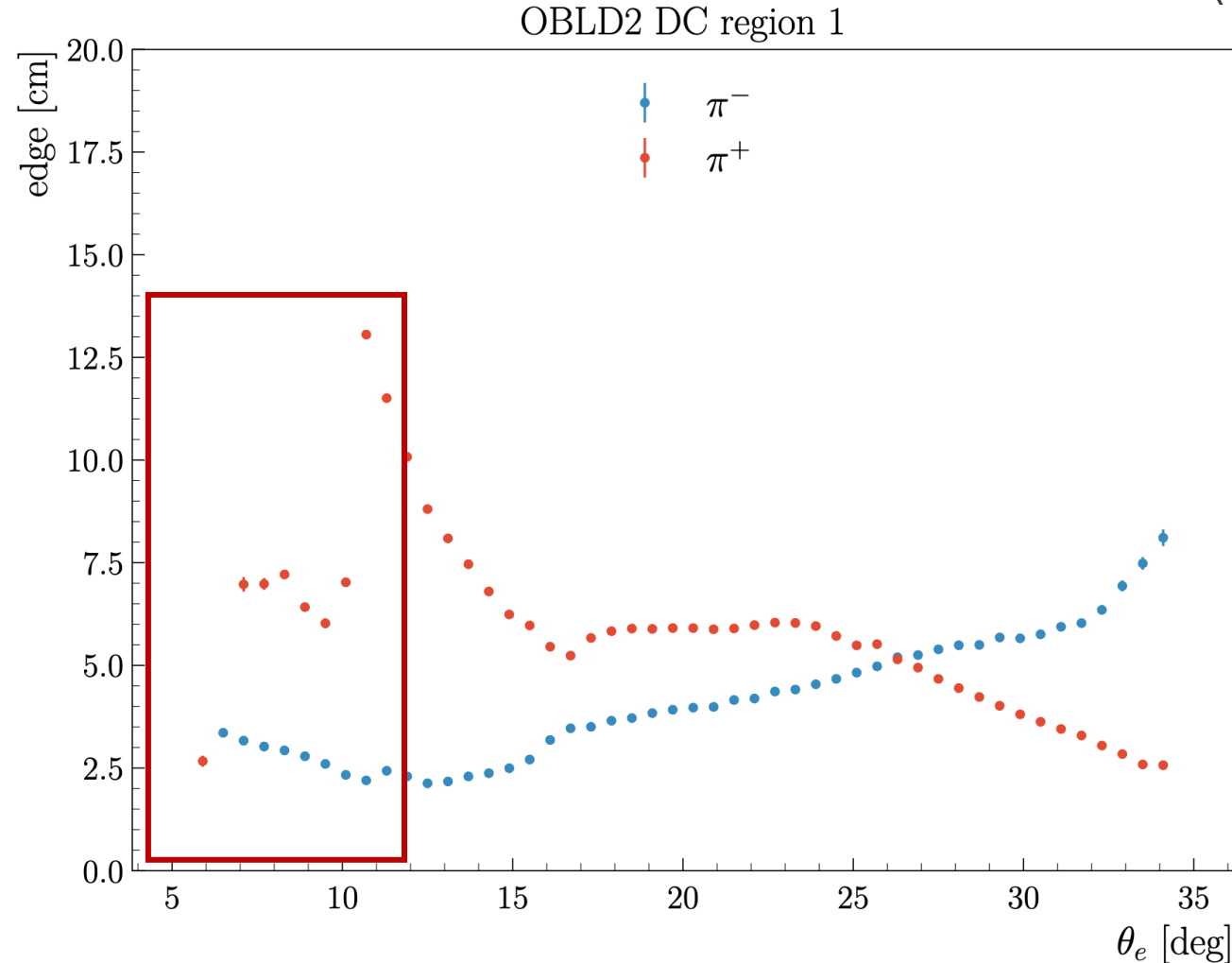


- Defining the “edge” limit
- Theta coordinates in the tilted-sector frame (all the sectors are on top of each other)
- Fitted with:
 - Gaussian form 0 to μ and line after

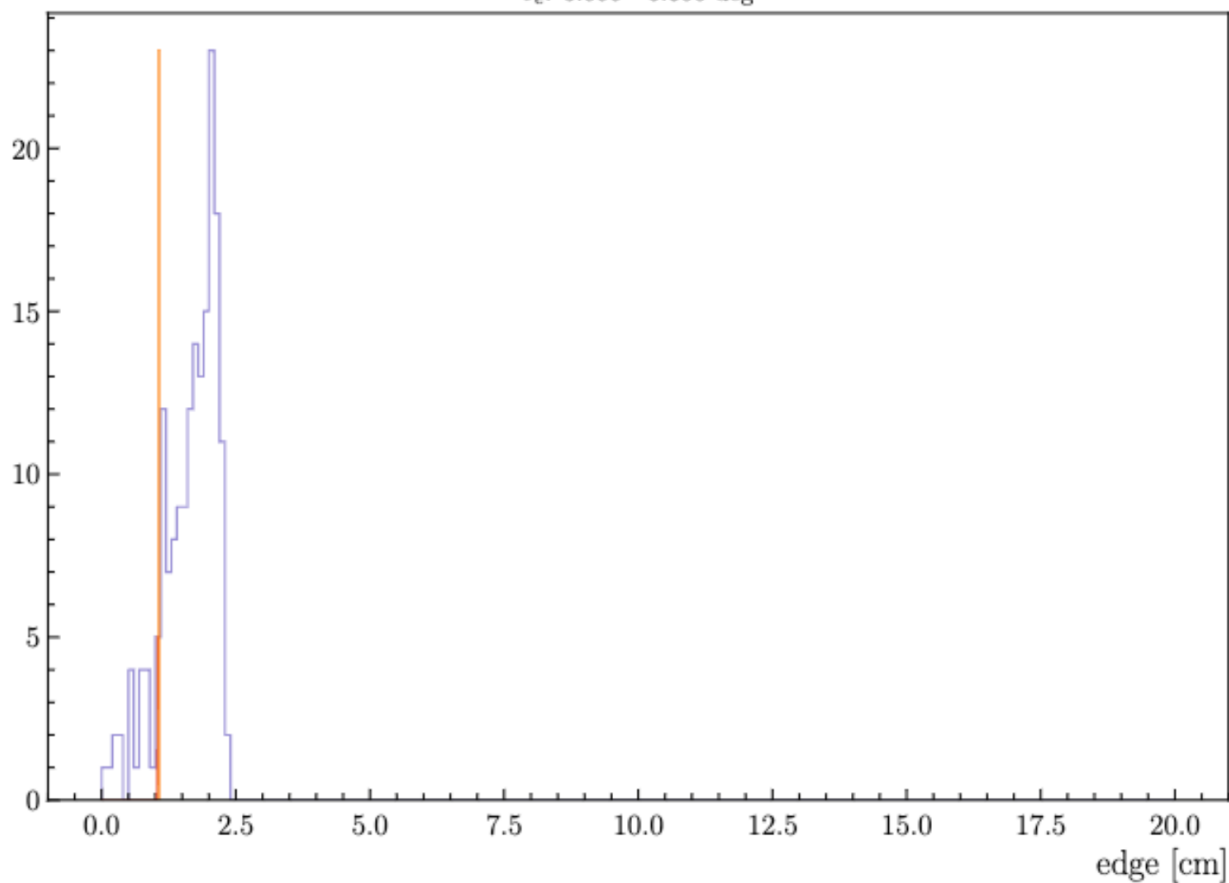
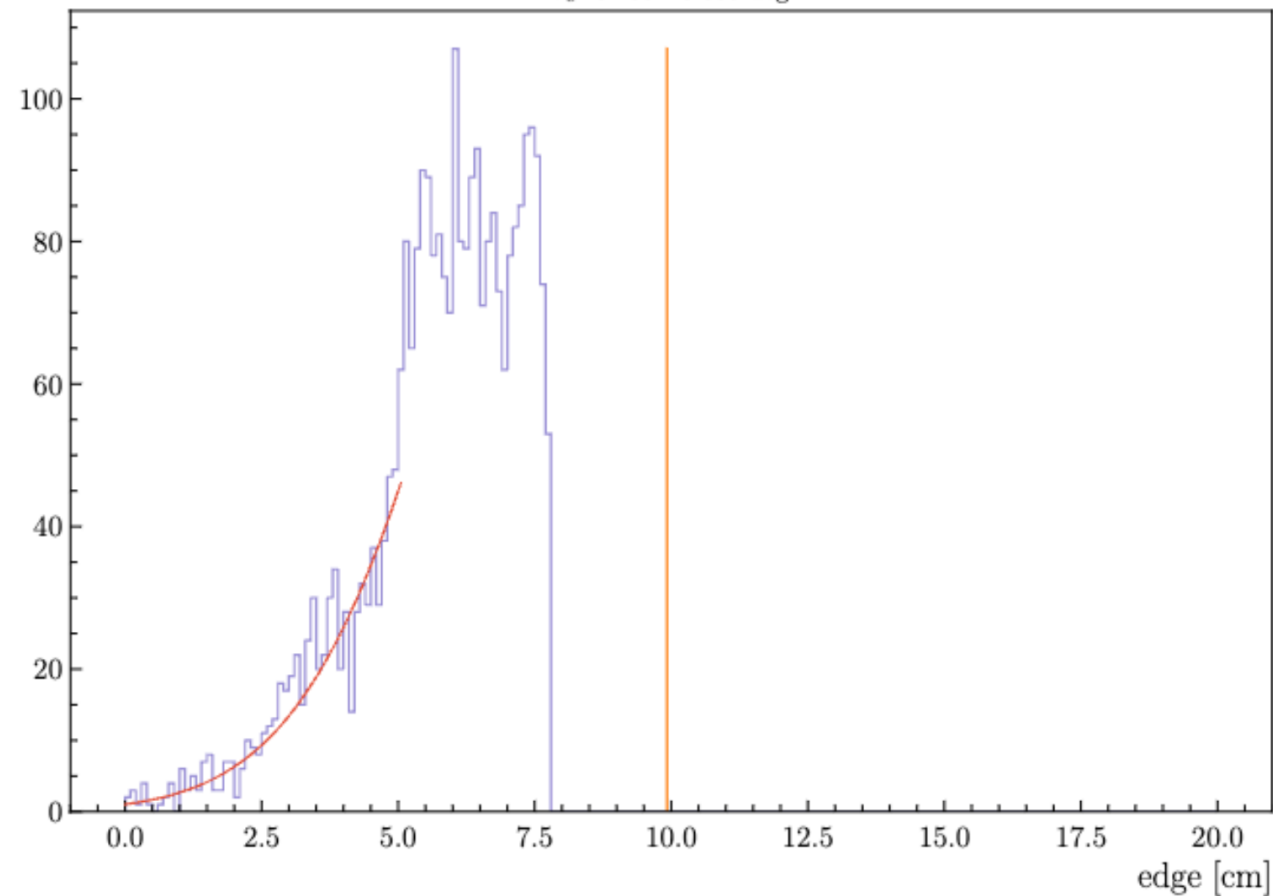
```
def fit_func(x: np.ndarray, pars: np.ndarray):  
    if np.abs(x[0]) < pars[1]:  
        return pars[0] * ROOT.TMath.Gaus(np.abs(x[0]), pars[1], pars[2])  
    else:  
        return pars[0] + (np.abs(x[0]) - pars[1]) * pars[3]
```



- Defining the “edge” limit
- Theta coordinates in the tilted-sector frame (all the sectors are on top of each other)



- Extract mu for each slices for both π^+ and π^-
- But I have some issues with the low theta.

$\chi^2/\text{ndf} = 3.33$
 $\theta_e: 5.000 - 5.600 \text{ deg}$

 $\chi^2/\text{ndf} = 1.52$
 $\theta_e: 6.200 - 6.800 \text{ deg}$


- I have fixed the simulation for ALERT:
- The issue was: the AHDC::adc bank in data has the leadingEdgeTime, timeOverThreshold, and constantFractionTime, but GEMC doesn't have them.
- These variables are filled from the AHDC::wf bank and the extractor that Felix has written.
- So, to fill them, I have added the extractor in COATJAVA.

```
{
  "name" : "AHDC::adc",
  "group": 22400,
  "item" : 11,
  "info": "ADC bank for the ALERT Wire Chamber",
  "entries":[
    { "name":"sector"      , "type":"B", "info":"sector 1"},
    { "name":"layer"      , "type":"B", "info":"layer 1-5"},
    { "name":"component"  , "type":"S", "info":"wire number"},
    { "name":"order"      , "type":"B", "info":"Stereo angle 0 - in, 1 - out"},
    { "name":"ADC"        , "type":"I", "info":"ADC maximum"},
    { "name":"time"       , "type":"F", "info":"time from pulse fit"},
    { "name":"ped"        , "type":"S", "info":"pedestal from pulse analysis"},
    { "name":"windex"     , "type":"S", "info":"row index in waveform bank"},
    { "name":"integral"   , "type":"I", "info":"sum of ADC"},
    { "name":"leadingEdgeTime" , "type":"F", "info":"time at constant fraction amplitude"},
    { "name":"timeOverThreshold" , "type":"F", "info":"time over threshold at constant fraction amplitude"},
    { "name":"constantFractionTime" , "type":"F", "info":"time from constant fraction discriminator"}
  ]
},
```

ALERT: Simulation

- I have fixed the simulation for ALERT:
- The issue was: the AHDC::adc bank in data has the leadingEdgeTime, timeOverThreshold, and constantFractionTime, but GEMC doesn't have them.
- These variables are filled from the AHDC::wf bank and the extractor that Felix has written.
- So, to fill them, I have added the extractor in COATJAVA.
- Added to AHDCEngine the line:

Fix: AHDC: use ahdcExtractor to process simulation #607



baltzell merged 2 commits into [JeffersonLab:development](#) from [mathieuouillon:development](#) 4 days ago

```
ahdcExtractor.update(30, null, event, "AHDC::wf", "AHDC::adc");
```

- I have also bypassed the criteria for a hit for simulation:

```
55 -         if ((adc >= adc_min) && (adc <= adc_max) && (leadingEdgeTime >= t_min) && (leadingEdgeTime <= t_max) && (timeOverThreshold >= tot_min) && (timeOverThreshold <= tot_max) && (adcOffset >= ped_min) &&
    (adcOffset <= ped_max)) {
57 +         if ((adc >= adc_min) && (adc <= adc_max) && (leadingEdgeTime >= t_min) && (leadingEdgeTime <= t_max) && (timeOverThreshold >= tot_min) && (timeOverThreshold <= tot_max) && (adcOffset >= ped_min)
    && (adcOffset <= ped_max)) || sim) {
```

- I have fixed the simulation for ALERT:
- For GEMC, after some discussion with Nathan, we chose to remove the AHDC::adc bank, so that we don't have to keep it updated (if someone changes the AHDC::adc bank again).

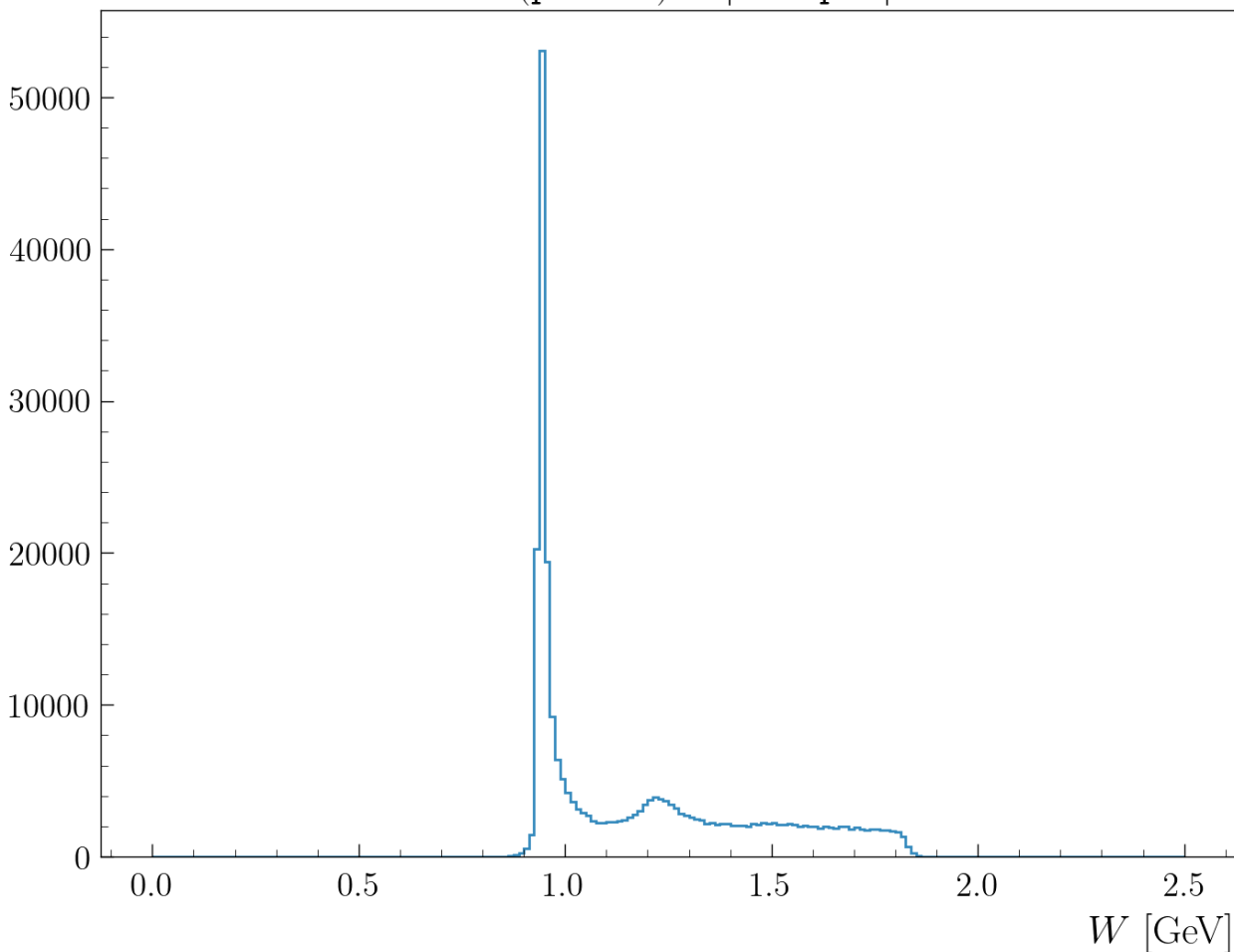
Remove the AHDC::adc bank #41

 Merged maureeungaro merged 3 commits into `gemc:main` from `mathieuouillon:main`  19 hours ago

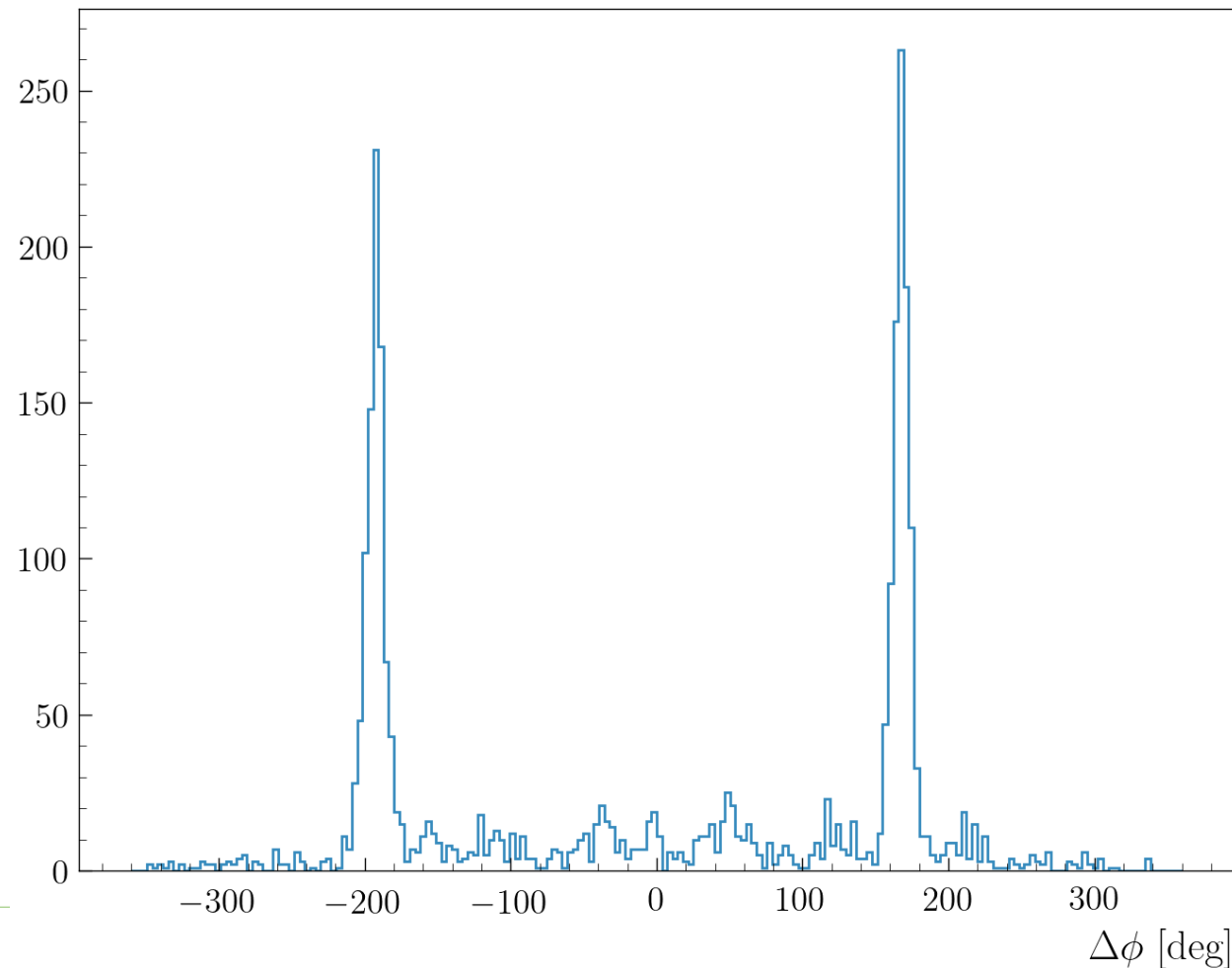
- All the pull requests have been accepted.

- I also work to test the AI on elastic data to estimate the efficiency and purity with real data.
- For that, I need to select the elastic electron from the data at 2 GeV on H2.

Electron ($\text{pid}=11$) + $|\text{chi2pid}| < 3$



ATOF hits



Task	expected	done	comment
Improve pid, fiducials cuts	04/21		Work on the DC fiducials, need to produce a PDF with all the plots and details, some issues with defining the DC limit
Produce physics plots for pass1 review	04/30		
Finish the event mixing method	05/04		
Extract the multiplicity ratio, pT	05/15		
improve code that generated all track candidates (ALERT)	05/15		
Test the AI on real data (ALERT)	04/21		Simulation work, new network have been trained, need to work on elastic data, new network have been train
Write the AI study on the wiki RG-D	05/12		

- Gabriel Charles organizes a workshop about gaseous detectors in France
- In June 18th to 20th
- He asked me if I could give a presentation about the AI-assisted track finding algorithm that has been developed for ALERT.
- The talk will be on Zoom, and there is no fee.